



Solutions Engineer – Technical Project

Thank you for taking the time to speak with us. We have enjoyed the interviews so far!

Our technical assignment is designed to test your practical skills, give you an idea of the work we do, and let us see how you explain complex topics in writing. Please note that completing this assignment is required before your next interview, which is the Technical Panel Round. In this round, you'll meet with our Solutions Engineering Team members. You'll present your assignment work and do a brief demo of the setup you created.

First steps:

1. Please add a domain (new or existing, e.g. yourwebsite.com) to Cloudflare. Cloudflare Free plan is sufficient.
2. Activate on Cloudflare by following the steps to change the nameservers at your DNS Registrar.

Please keep in mind that while we can help point you in the right direction, this assignment is meant to evaluate your ability to learn new concepts quickly through effective research & self-discovery. The Cloudflare offering is vast, and it's crucial that you are comfortable researching answers on topics you may not be familiar with. In case you have any questions, technical clarifications or troubleshooting help needed – please reach out to your Cloudflare Recruiter and we will get back to you as soon as we can.

Additional considerations:

- Keep the audience in mind: your reviewers are technical experts on Cloudflare, but also want to understand how you can communicate complex topics to non-technical stakeholders.
- Technical deliverables are largely mutually exclusive. If you have trouble with any of the specific tasks this should not deter or prevent you from completing the remainder of the tasks.
- You will need to be ready to discuss your homework in the next round of interviews. Please submit your homework to your friendly Cloudflare Recruiter in a PPT/Slide format – that you will also be presenting on the day of the Technical Panel Round.



Steps:

1. Create an origin web server on a platform of your choosing. This could be in AWS, Google Cloud, DigitalOcean, your Raspberry Pi, etc.

This web server must run an endpoint that returns all HTTP request headers in the body of the HTTP response.

The web server can be something that you have written yourself (e.g. in JavaScript, Python, etc) or by using a 3rd party application.

2. Proxy traffic to this server through Cloudflare. Add necessary configurations on Cloudflare. 3. Secure the communication between Cloudflare and your Origin Server with a non-Cloudflare provisioned TLS certificate using at least the Full-Strict mode on Cloudflare.

4. Set a rate limiting rule on Cloudflare. How would you demonstrate to a customer that this rate limiting rule works?

5. Install and configure Cloudflare Tunnel on your origin server using a subdomain called "tunnel", e.g. tunnel.yourwebsite.com. Make connections proxied to your server protected using this tunnel.

6. Configure an SSO IdP (Identity Provider) of your choosing within **Cloudflare Zero Trust**. Please note that Cloudflare also provides an "OTP" option, in case you do not have any IdPs available to configure.

7. Lock down access for a particular path for your Cloudflare Tunnel subdomain (e.g. tunnel.yourwebsite.com/secure) and only allow access for yourself (with the previously configured IdP of your choosing) and users with an @cloudflare.com email address .

- a. Ensure nobody can bypass Cloudflare and access your server's IP directly.

8. Create a Cloudflare Worker.

Serve a Worker on the tunnel.yourwebsite.com/secure path that returns identity information for the authenticated user.

The HTTP response body should contain: "\${EMAIL} authenticated at \${TIMESTAMP} from \${COUNTRY}"

\${COUNTRY} should be an HTML link that when clicked navigates to tunnel.yourwebsite.com/secure/\${COUNTRY} and displays the appropriate country flag.

The flag asset should be stored in a private R2 bucket.



- a. Create this Worker using the Wrangler CLI, upload your Workers code to a public Git repository for your implementation.
- b. The `/secure` response should be returned as HTML
- c. The `/secure/${COUNTRY}` response should use an appropriate content type

Deliverables:

1. A working application with instructions needed to access any of the above.
2. A report (PPT/Slide format) that addresses the following:
 - A report detailing the requirements and how you implemented them step by step.
 - What use cases can you see different products being useful for?
 - How did you fill the gaps (if any) in your knowledge during the process?
 - How do you imagine that a target customer will find this experience?

Please let us know if you have any questions with any of the above. We wish you all the best for the assignment and hope that you have fun as you go through this journey together with us!